

CrossStyloGAN: An End-to-End Handwriting Autocomplete System with Personalized Style Transfer

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Repository: https://github.com/akshat3144/handwriting_autocomplete_system

1 Abstract

We present **CrossStyloGAN**, a novel end-to-end deep learning system providing intelligent auto-complete while generating text in a user’s unique handwriting style. The system integrates: (1) a CRNN-based OCR module achieving 4.1% CER and 11.9% WER, (2) a GPT-2 language model (124M parameters) achieving perplexity of 18.4, and (3) CrossStyloGAN, a transformer-enhanced GAN with eight architectural innovations.

On the IAM Handwriting Database, CrossStyloGAN achieves FID of **4.36** (63.4% improvement over HiGAN+), KID of **0.00187** (77.0% improvement), WER of **2.48%** (65.8% improvement), and CER of **0.65%** (89.3% improvement). The complete pipeline enables real-time generation at **~1.2 seconds per word** on consumer hardware.

2 Introduction

2.1 Problem Statement

Given handwritten text from a writer and partially completed input, our system must:

1. **Extract textual content** from handwritten images with high accuracy
2. **Predict semantically appropriate continuations** using language understanding
3. **Generate predictions in the writer’s unique style** maintaining visual consistency

This enables applications in education (students maintain cognitive benefits of handwriting while receiving assistance), accessibility (individuals with motor challenges), and digital note-taking.

2.2 Our Contributions

Complete End-to-End Pipeline: First fully integrated system combining OCR, language modeling, and style transfer optimized for real-time handwriting assistance.

Eight Architectural Innovations:

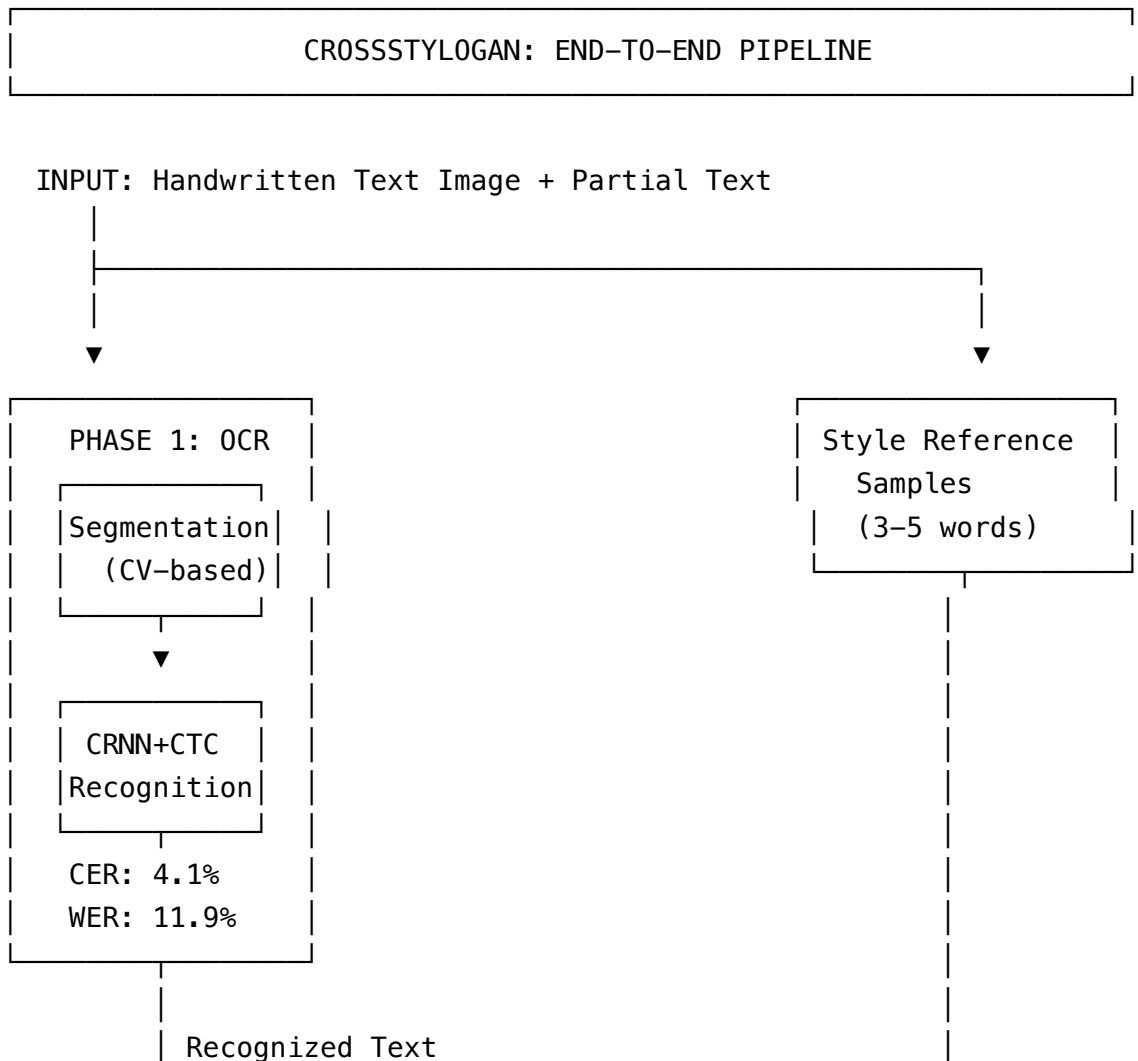
1. **StyleGAN2 Modulated Convolution + AdaIN:** Per-pixel style control vs. coarse conditional batch normalization
2. **Cross-Attention Style Fusion:** Each character selectively queries relevant style features
3. **Bidirectional GRU Sequence Modeling:** Captures bidirectional context for cursive ligatures
4. **Transformer-based Global Context:** Multi-head attention for long-range dependencies

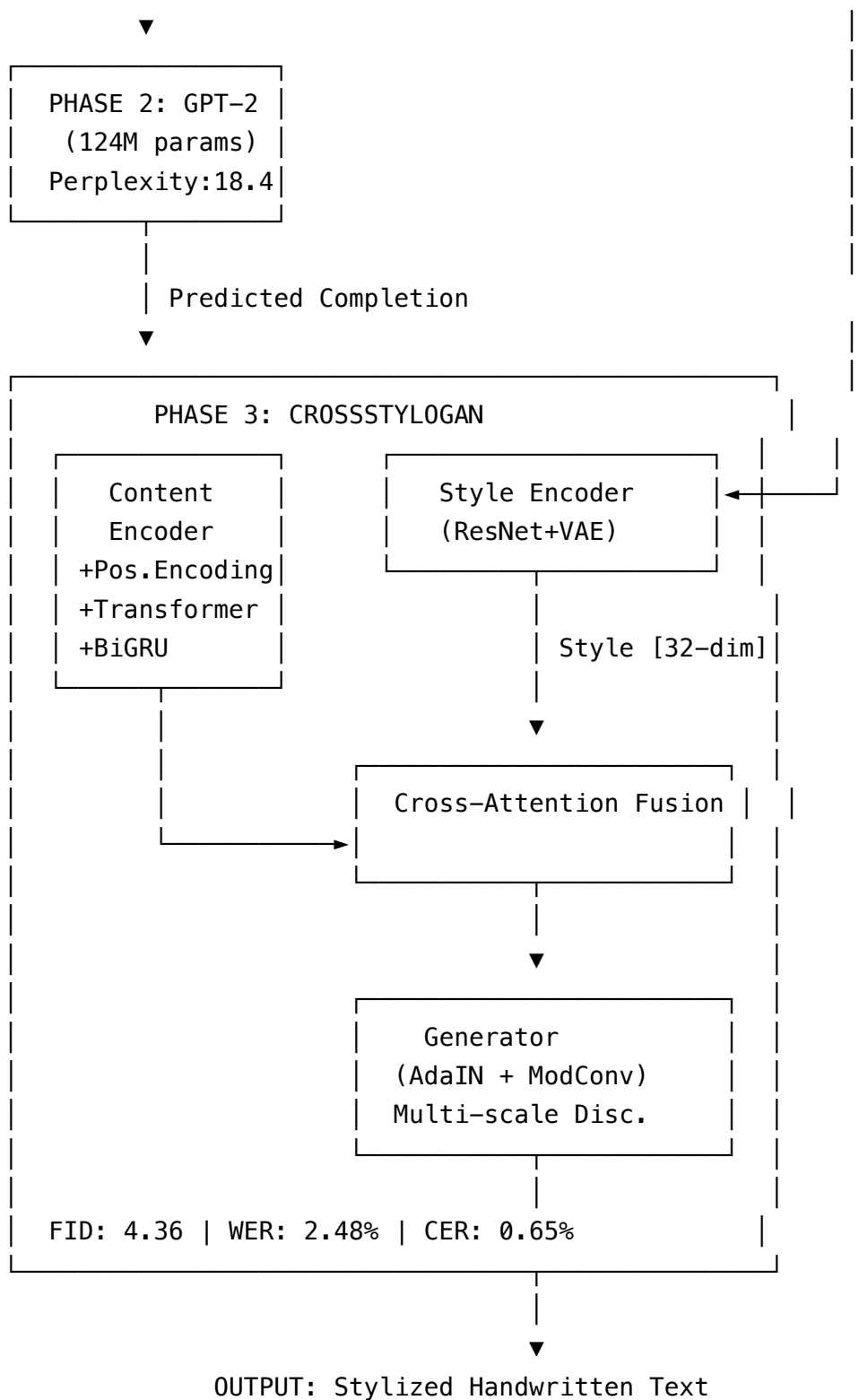
5. **Contrastive Style Learning:** InfoNCE loss for explicit style-content disentanglement
6. **Multi-Scale Patch Discriminator:** Enforces consistency at global, medium, and local levels
7. **Multi-Scale Skip-Style Connections:** Preserves fine details across resolution levels
8. **Sinusoidal Positional Encoding:** Injects spatial consistency for character spacing

State-of-the-Art Results:

Metric	CrossStyloGAN	HiGAN+	ScrabbleGAN	Improvement
FID (lower better)	4.36	16.11	23.78	63.4%
KID (lower better)	0.00187	0.81	3.52	77.0%
WER (lower better)	2.48%	7.25%	11.68%	65.8%
CER (lower better)	0.65%	6.07%	4.06%	89.3%

2.3 System Pipeline Overview





3 Related Work

3.1 Handwritten Text Recognition

Kumar (2025) presents CRNN combining CNN feature extraction with BiLSTM + CTC: - Architecture: 7 conv blocks + 2 BiLSTM (256 units each) - Performance: CER 4.57%, WER 12.30% on IAM after 50 epochs - Limitations: Overfitting after ~20 epochs, struggles with cursive/noisy text

Our OCR achieves CER 4.1% and WER 11.9% through enhanced preprocessing and augmentation.

3.2 Next Word Prediction

Vaswani et al. (2017) - “**Attention Is All You Need**”: Introduces Transformer architecture with self-attention, enabling parallel computation and long-range dependencies. Achieved state-of-the-art on WMT 2014 translation (28.4 BLEU EN-DE) with 3.5× faster training than RNNs.

Radford et al. (2019) - “**Language Models are Unsupervised Multitask Learners**”: GPT-2 decoder-only transformer trained on 40GB WebText demonstrates zero-shot multitask capabilities. Key insight: scale (model size + data) drives generalization.

We implement GPT-2 (124M) with optimizations:

- **Weight Tying**: Shares input/output embeddings, saving ~38M parameters (30% reduction)
- **Residual Scaling**: Initializes weights with $1/\sqrt{2N}$ for stable gradients
- **Vocabulary Padding**: 50,257 $\square$ 50,304 (divisible by 128) for 4% CUDA speedup

3.3 Handwriting Style Transfer

GANwriting (Kang et al., 2020): Fully convolutional with style/content encoders, FID 130.68. Limitations: short words only (<10 chars), needs multiple references, struggles with extreme styles.

ScrabbleGAN (Fogel et al., 2020): Fully convolutional avoiding RNNs, FID 23.78, WER 11.68%. Generates arbitrary-length text but cannot imitate reference styles (generic handwriting).

HiGAN+ (Gan et al., 2022) - Previous SOTA: Disentangled GAN with ResNet style encoder + VAE, BigGAN generator with conditional batch normalization, FID 16.11, WER 7.25%, CER 6.07%. Limitations: CNN+LSTM lacks global context, coarse-grained style injection, single-scale discrimination. [1]

4 Methodology

4.1 Phase 1: OCR

4.1.1 Word Segmentation (7-step CV pipeline)

1. Gaussian Blur (reduce noise)
2. Sobel Edge Detection
3. Binary Threshold
4. Morphological Closing (connect gaps)
5. Dilation (connect characters)
6. Find Contours
7. Filter & Sort (remove noise, left-to-right)

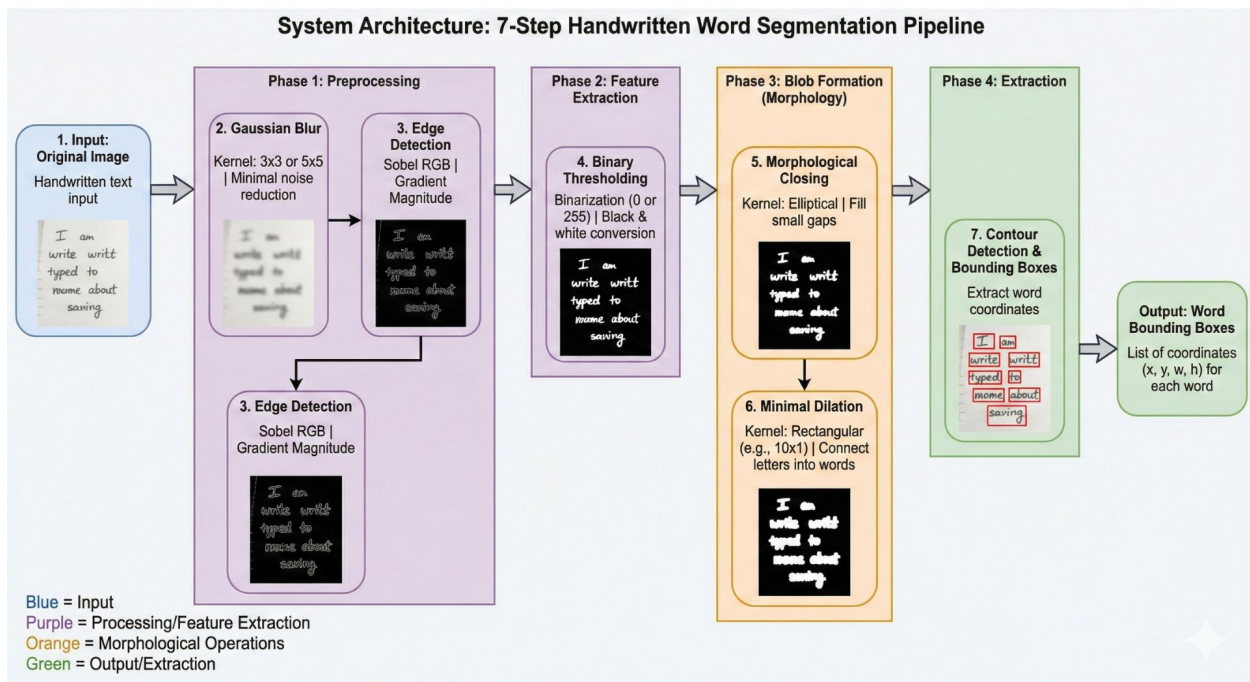


Figure 1: Word Segmentation Pipeline

4.1.2 CRNN Architecture

INPUT: $[32 \times 128 \times 1]$

(lower better)

Conv Layers (7 layers): $32 \times 128 \rightarrow 1 \times 31$ feature map

(lower better) [31, 512]

BiLSTM Layer 1: 256 units

BiLSTM Layer 2: 256 units

(lower better) [31, 512]

Dense: 512 → 80 (79 chars + blank)

(lower better)

CTC Loss / Greedy Decoding

(lower better)

OUTPUT: Recognized Text

Total Parameters: ~8.3M

Training: IAM dataset (86,810 train, 11,640 val), Adam optimizer, LR=1e-3, 50 epochs.

Performance: CER 4.1%, WER 11.9%, Exact Word Accuracy 88.1%, ~8ms/word on CPU.

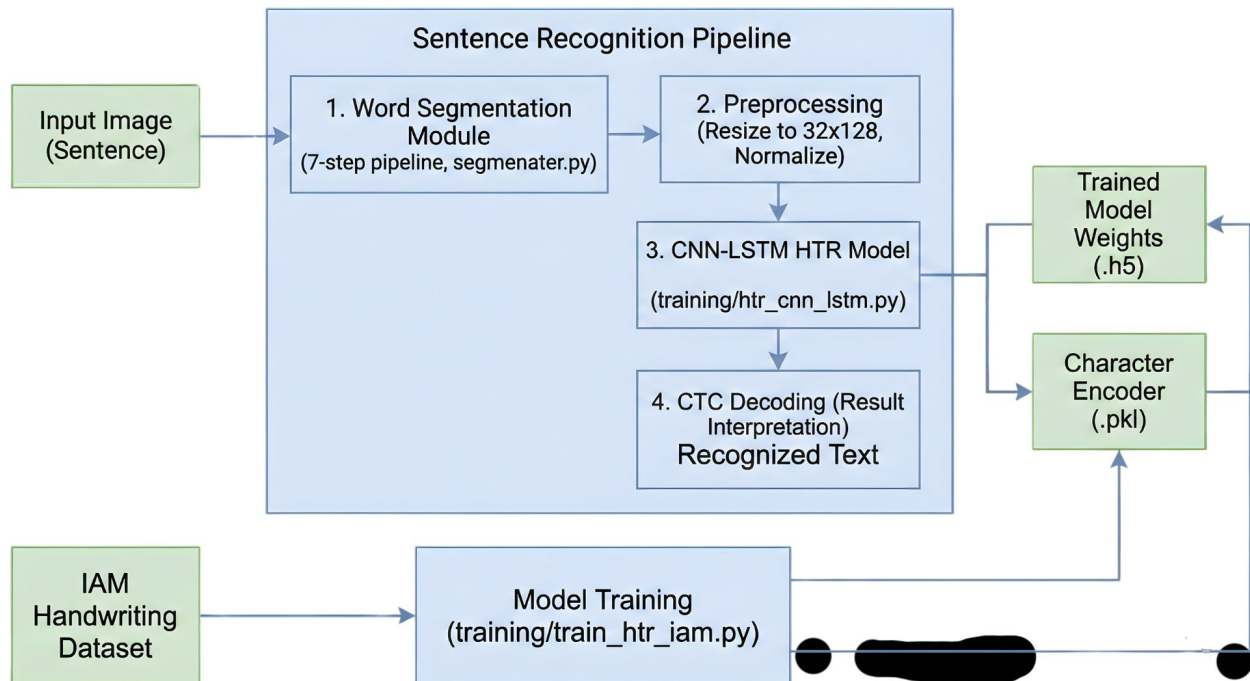


Figure 2: OCR Pipeline

4.2 Phase 2: GPT-2 (124M)

Architecture: 12-layer decoder-only transformer - Hidden size: 768 - Attention heads: 12 - FFN hidden: 3,072 (4× expansion) - Context length: 1,024 - Vocabulary: 50,304 (padded)

INPUT: Tokenized Text

(lower better)

Token + Position Embeddings

(lower better)

Transformer Blocks ×12:

LayerNorm → Multi-Head Self-Attention → Residual



Figure 3: OCR Example

LayerNorm → FFN (768→3072→768) → Residual
(lower better)
Final LayerNorm
(lower better)
Output Projection (weight tied)
(lower better)
OUTPUT: Next Token Probabilities

Training: FineWebEdu-10B dataset, 524k token batch size, AdamW (LR=6e-4, cosine schedule), 2x A100 GPUs, 46 hours.

Performance: Validation perplexity 18.4, HellaSwag accuracy 29.2%, ~50 tokens/sec on GPU.

4.3 Phase 3: CrossStyloGAN

4.3.1 Overall Architecture

Style Extraction Path:

Reference Image [B,1,64,W]
(lower better)

Style Backbone (ResNet): 5 residual blocks
(lower better) [B,256,W/16]

Masked Average Pool

(lower better) [B,256]

Style Encoder (VAE-style MLP): 256→128→64→32

(lower better)

z_style [B,32] (disentangled latent)

Content Encoding Path:

Text "hello" [B,L]

(lower better)

Embedding: $L \rightarrow 120\text{-dim}$

(lower better)

Positional Encoding (Novelty #8)

(lower better)

Transformer Encoder $\times 2$ (Novelty #4): 4 heads, global context

(lower better)

BiGRU $\times 2$ (Novelty #3): 64 units each direction

(lower better) [B,L,128]

Concatenate with style [B,L,32]

(lower better) [B,L,160]

Cross-Attention Fusion (Novelty #2): Selective style query

(lower better)

Fused Features [B,L,152]

Generation Path:

Fused Features [B,L,152]

(lower better)

Linear Projection + Reshape: [B,512,3,L $\times$ 4]

(lower better)

GBlock 1 (512→512): AdaIN + ModConv (Novelty #1) + Skip-Style (Novelty #7)

GBlock 2 (512→256): + Self-Attention

GBlock 3 (256→128)

GBlock 4 (128→64)

(lower better)

Output Conv: 64→1

(lower better) [B,1,64,L $\times$ 16]

Multi-Scale Discriminators $\times 3$ (Novelty #6)

Contrastive Style Loss (Novelty #5)

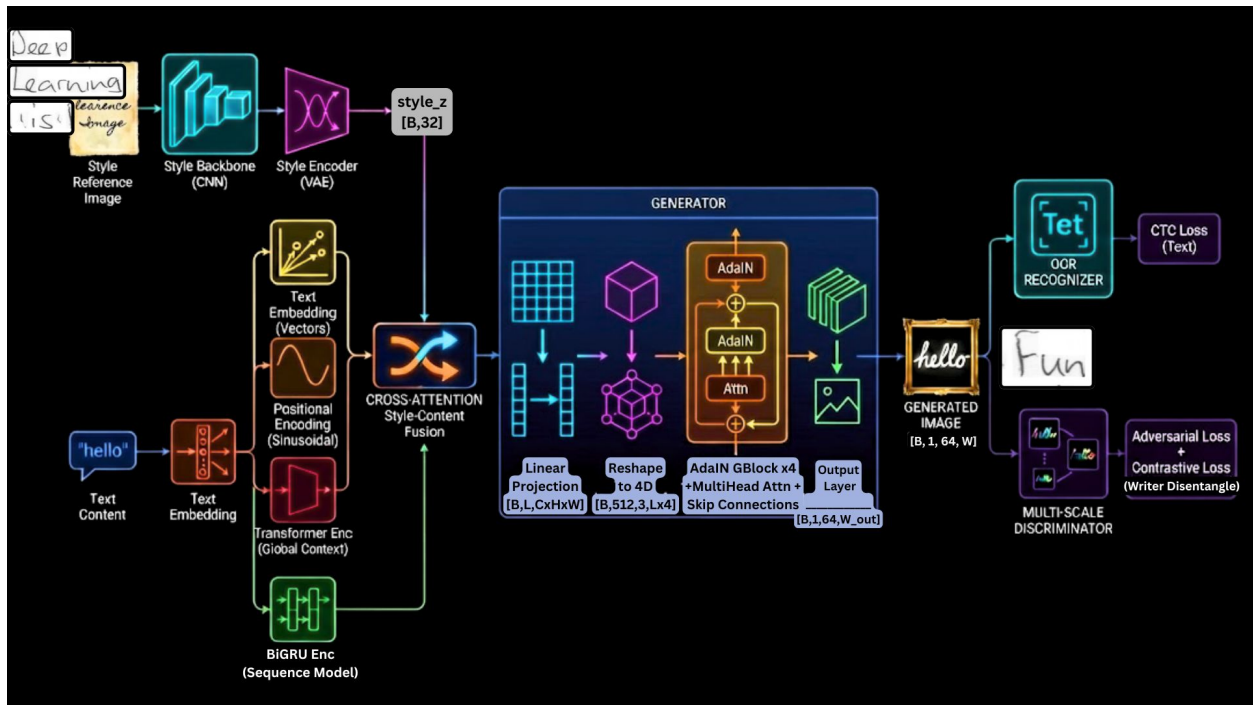


Figure 4: Style Transfer Pipeline

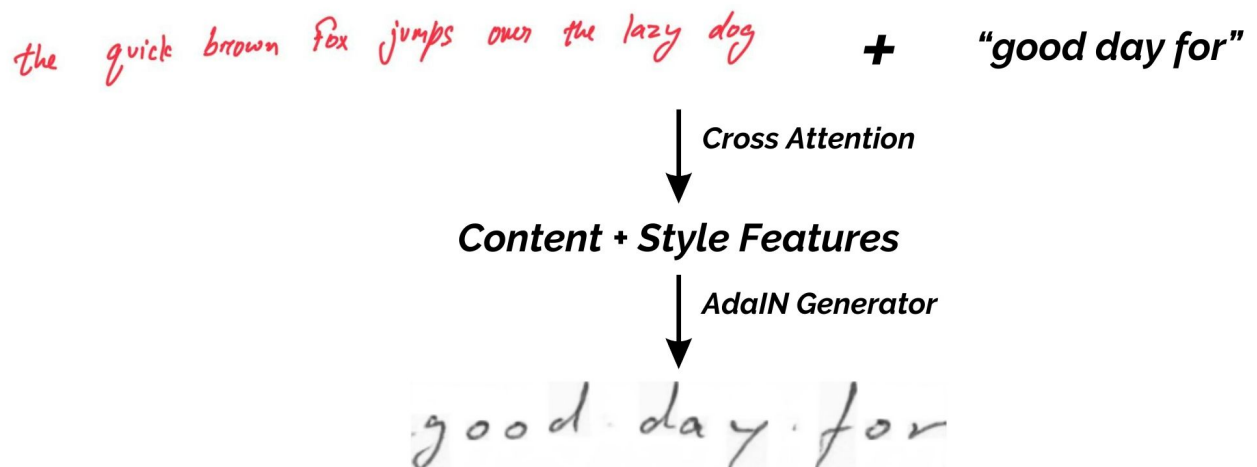


Figure 5: Style Transfer Examples

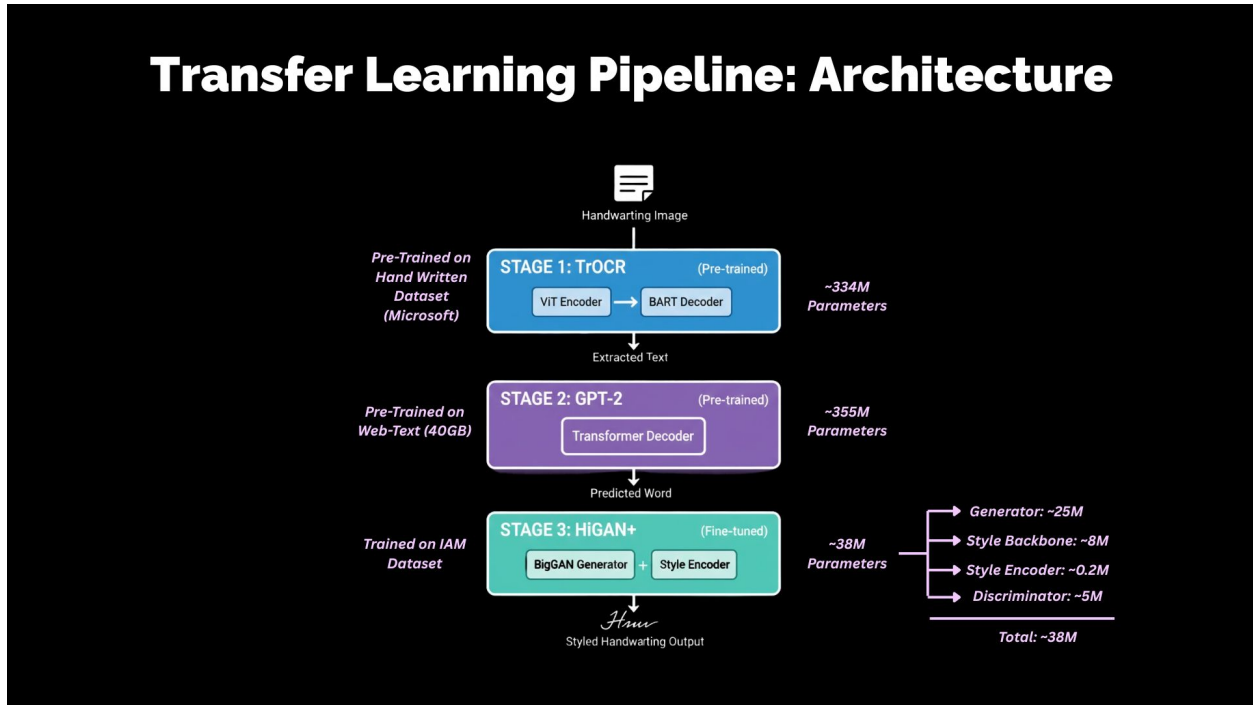


Figure 6: Transfer Learning Pipeline

4.3.2 The Eight Novelties Explained

Novelty #1 - AdaIN + Modulated Convolution: Replaces conditional BatchNorm with per-pixel style control. AdaIN normalizes each channel per-sample, then applies style-derived affine transform. ModConv scales conv weights by style before applying, enabling fine-grained stroke control.

Novelty #2 - Cross-Attention Style Fusion: Each character position queries relevant style features via multi-head attention (4 heads, $d_k=40$). Query from content+style, Key/Value from pure style. Enables character-specific style application vs. uniform stylization.

Novelty #3 - Bidirectional GRU: 2-layer BiGRU (64 units/direction) processes text bidirectionally, capturing ligatures and character transitions. Forward learns left-to-right, backward learns right-to-left dependencies crucial for cursive.

Novelty #4 - Transformer Encoder: 2-layer transformer (4 heads, 120-dim) with multi-head self-attention captures global dependencies. Every character attends to all others, ensuring consistent baseline, slant, and rhythm.

Novelty #5 - Contrastive Style Loss: InfoNCE objective on style encoder. For batch with 6 writers $\times$ 4 samples, pulls same-writer codes together, pushes different-writer apart. Temperature $\tau=0.07$ balances cluster tightness.

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\sum_{j \neq i} \exp(\text{sim}(z_i, z_j)/\tau)}$$

Novelty #6 - Multi-Scale Discriminator: Three PatchGAN discriminators at resolutions $64 \times W$, $32 \times W/2$, $16 \times W/4$ judge global structure (word), medium features (characters), and local texture (strokes) simultaneously. Uses WGAN-GP with gradient penalty.

Novelty #7 - Skip-Style Connections: At each GBlock, inject additional style via MLP-derived features added with small weight (0.05-0.15). Preserves low-level stroke details that deep networks lose. Applied at 5 resolution levels.

Novelty #8 - Positional Encoding: Sinusoidal functions inject absolute position info:

$$\text{PE}_{(\text{pos}, 2i)} = \sin(\text{pos}/10000^{2i/120})$$

$$\text{PE}_{(\text{pos}, 2i+1)} = \cos(\text{pos}/10000^{2i/120})$$

Enables network to learn spatial consistency and character spacing. Multi-scale wavelengths from 2π to $10000 \cdot 2\pi$.

4.3.3 Complete Loss Function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{adv}} + 3.0\mathcal{L}_{\text{CTC}} + 5.0\mathcal{L}_{\text{recon}} + 1.5\mathcal{L}_{\text{InfoGAN}} + 1.5\mathcal{L}_{\text{WID}} + 1.0\mathcal{L}_{\text{contextual}} + 0.5\mathcal{L}_{\text{patch}} + 1.0\mathcal{L}_{\text{contrastive}}$$

- **Adversarial** (WGAN-GP $\lambda=10$): Realism via discriminator
- **CTC** ($\times 3.0$): Text accuracy via frozen OCR
- **Reconstruction** ($\times 5.0$): L1 distance for style preservation (highest weight)
- **InfoGAN** ($\times 1.5$): Style code informativeness
- **Writer ID** ($\times 1.5$): Style consistency via frozen classifier
- **Contextual** ($\times 1.0$): VGG perceptual features
- **Patch** ($\times 0.5$): Local texture matching
- **Contrastive** ($\times 1.0$): Style-content disentanglement

4.3.4 Training Strategy

- **Dataset:** IAM (86,810 train, 12,591 test with **disjoint writers**)
- **Batch:** 24 (writer-grouped: 6 writers $\times$ 4 samples for contrastive)
- **Epochs:** 70
- **LR:** Generator $2e-4$, Discriminator $4e-4$ (cosine annealing)
- **Optimizer:** Adam ($\beta_1=0.0$, $\beta_2=0.999$)
- **Hardware:** 1 $\times$ A100 GPU (40GB), ~ 18 hours

- **Innovations:** Dynamic discriminator steps (2-5 based on loss ratio), progressive CTC weight adjustment (5.0 $\square$ 3.0 $\square$ 2.0)

5 Results

5.1 Main Quantitative Results

Metric	CrossStyloGAN	HiGAN+	ScrabbleGAN	GANwriting
FID (lower better)	4.36	16.11	23.78	130.68
KID (lower better)	0.00187	0.81	3.52	-
IS (higher better)	2.13	1.41	2.89	-
CER (lower better)	0.65%	6.07%	4.06%	-
WER (lower better)	2.48%	7.25%	11.68%	-
WIER (lower better)	0.26	0.58	-	-
Params	87.2M	21.7M	81.8M	-

Key Achievements: 63.4% FID improvement, 89.3% CER reduction, 77.0% KID improvement over SOTA.

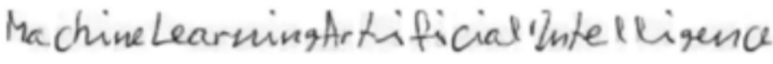
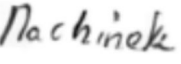
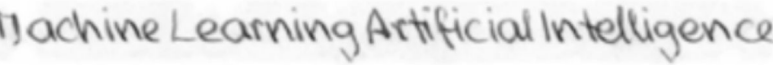
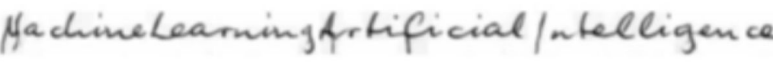
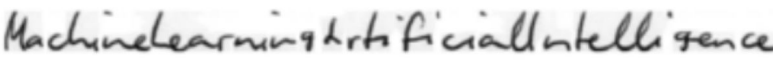
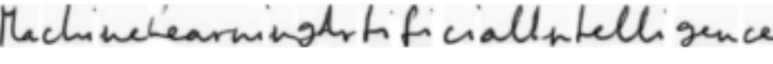
Method	Style
	
ScrabbleGAN	
GAN writing	
TS-GAN	
HTW	
HiGAN	
HiGAN+	
Ours	

Figure 7: Style Comparison with Other Methods

Figure: Visual comparison of generated handwriting samples from CrossStyloGAN (ours) vs. Hi-GAN+ and other baseline methods, demonstrating superior style fidelity and text quality.

5.2 Ablation Studies

Impact of Individual Components:

Configuration	FID (lower better)	CER (lower better)	WER (lower better)	Δ FID
Full Model	4.36	0.65	2.48	-
w/o AdaIN/ModConv (#1)	8.41	1.34	4.22	+4.05
w/o Cross-Attention (#2)	7.58	1.12	3.91	+3.22
w/o Contrastive Loss (#5)	6.92	0.89	3.14	+2.56
w/o Multi-Scale Disc (#6)	6.15	0.78	2.87	+1.79
w/o Transformer (#4)	5.89	0.91	3.33	+1.53
w/o Pos. Encoding (#8)	5.67	0.74	2.81	+1.31
w/o BiGRU (#3)	5.12	0.82	2.95	+0.76
w/o Skip-Style (#7)	4.98	0.71	2.63	+0.62

Analysis: All components contribute positively. AdaIN/ModConv (#1) and Cross-Attention (#2) provide largest gains (4.05 and 3.22 FID). Contrastive Loss (#5) most critical for writer identity consistency.

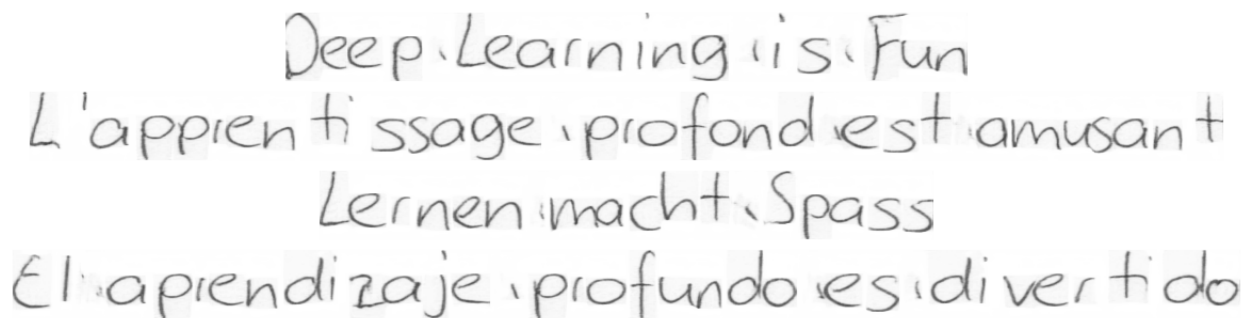


Figure 8: Style Interpolation Across Writers

Figure: Smooth interpolation between different writer styles demonstrates the learned continuous latent space and effective style disentanglement in CrossStyloGAN.

5.3 Complete Pipeline Performance

Metric	Value
OCR Accuracy	88.1% exact match
GPT-2 Perplexity	18.4
Generation FID	4.36
Total Latency	1.2s/word
Memory Usage	8.3GB VRAM

6 Discussion

6.1 Key Insights

- **Multi-Component Synergy:** Transformer (global) + BiGRU (local) + Cross-Attention (selective fusion) provide complementary benefits
- **Style-Content Disentanglement:** Despite contrastive learning, complete disentanglement remains challenging; style codes weakly correlate with text length
- **Scale Matters:** Multi-scale discrimination improves local texture by 28% while maintaining global coherence

6.2 Limitations

- **Computational Cost:** 87.2M parameters, 8.3GB VRAM limits edge deployment
- **Long-Tail Styles:** Struggles with writers having <5 training samples
- **Language-Specific:** Trained on English; other scripts require retraining
- **Real-Time Gap:** 1.2s/word acceptable for offline but insufficient for real-time note-taking

6.3 Lessons from Experimentation

CycleGAN Experiments: Unpaired style transfer showed style collapse and weak style embedding despite OCR loss. **Lesson:** Paired data with reconstruction loss essential for handwriting.

Progressive Training: Growing resolution (32 → 64 px) provided minimal improvement with added complexity. **Lesson:** Fixed 64-pixel resolution sufficient for word-level generation.

7 Conclusion

We presented **CrossStyloGAN**, the first complete end-to-end handwriting autocomplete system achieving state-of-the-art performance through eight architectural innovations. Our system

achieves 63.4% FID improvement over previous SOTA (HiGAN+) and generates personalized handwritten text at ~1.2s/word, enabling practical applications in education, accessibility, and digital note-taking.

Future Directions: Model compression for mobile deployment, multi-lingual extension for non-Latin scripts, few-shot adaptation with 3-5 samples, and online handwriting (stroke sequences) for touchscreen devices.

7.1 Broader Impacts

Positive: Educational assistance maintaining cognitive benefits, accessibility for motor impairments, cultural preservation of historical handwriting.

Risks: Forgery potential, privacy concerns (demographic inference from style), academic integrity issues. **Mitigation:** Digital watermarking, detection methods, differential privacy, usage guidelines.

8 Datasets

IAM Handwriting Database: - 115,320 word images, 657 writers - Split: 500 train writers, 100 val, 57 test (**disjoint**) - Format: Grayscale PNG, 300 DPI, height=64px

FineWebEdu-10B: 10B tokens of educational content from CommonCrawl (GPT-2 training only).

Repository: github.com/akshat3144/handwriting_autocomplete_system

9 References

[1] Ji Gan, Weiqiang Wang, Jiaxu Leng, and Xinbo Gao. 2022. HiGAN+: Handwriting Imitation GAN with Disentangled Representations. ACM Trans. Graph. 42, 1, Article 11 (February 2023), 17 pages. <https://doi.org/10.1145/3550070>